ITRI625 — NETWORK SECURITY

Study Unit 2: Database & Big Data Security

0–5 MIN · SETTING THE STAGE

SLIDE 1

Title Slide · *suggested time ~0:30*

SCRIPT

Good morning/afternoon, everyone. Welcome to Study Unit 2 of Network Security. Today's session is on Database and Big Data Security,

DELIVERY CUE

Stand centre-stage while the title is up. Don't start talking the second the slide appears — let people finish reading it, then begin. Make eye contact across the room, not just the front row.

SLIDE 2

Learning Outcomes · *suggested time ~1 min*

SCRIPT

By the end of this session you should be able to: one, discuss how access control is implemented in databases; two, explain how sensitive data is protected; three, understand data mining and Big Data.

DELIVERY CUE

Point at each number as you say it (1, 2, 3) — this is a visual anchor students will subconsciously use to track the whole talk.

SLIDE 3

Context — Databases: A High-Value Target · *suggested time ~2 min*

SCRIPT

Let's set the stakes quickly. Databases hold an organisation's crown jewels — customer records, financial data, medical files. A breach here isn't abstract or theoretical: it means fraud, lawsuits, and regulatory fines. To put numbers on that: the estimated annual cost of data breaches in South Africa alone is R2.2 billion. Over 34.5 million South African accounts were breached in just the first quarter of 2024. And globally, the average cost of a single data breach in IBM's 2025 report is $4.44 million. That's the backdrop for everything we're about to cover — how do we actually stop this?

DELIVERY CUE

Pause briefly after each statistic — let the number land before moving to the next one. Don't read all three stats in one breath.

5–13 MIN · OUTCOME 1 — ACCESS CONTROL

SLIDE 4

Outcome 1 Divider · *suggested time ~20 sec*

SCRIPT

Outcome one: access control in databases. In short — how organisations decide who can see, change, or delete data, and how they prove it afterwards.

Access Control & Auditability · *suggested time ~2.5 min*

SCRIPT

Access control implementation rests on three principles. Confidentiality — only authorised users can reach sensitive fields. Integrity — only permitted users can change data. And least privilege — users get the minimum access their role actually needs, nothing more. On the practical side, that means: define roles before you write any permissions; grant least-privilege access by default; separate duties so no single user can do everything alone; enforce MFA on privileged accounts; log and review access to sensitive tables; and revoke access immediately the moment someone's role changes. That last point connects to auditability — logs of who did what and when. Without that, you can't prove compliance with something like POPIA, and you can't detect suspicious activity after the fact.

DELIVERY CUE

This slide has the most content on the page — don't read the checklist line by line. Group it into the two ideas (what access control does / how you make it auditable) and speak in full sentences, using the bullets only as your own memory prompt.

SLIDE 6

Three Access Control Models — DAC, MAC, RBAC · *suggested time ~2 min*

SCRIPT

There are three classic models for implementing this. DAC — Discretionary Access Control — the data owner decides who gets access. Think of a lecturer sharing a spreadsheet with chosen students: flexible, but it relies entirely on individual judgement. MAC — Mandatory Access Control — access is fixed by a classification level set centrally, and the user cannot override it. Military and government systems are the classic example. And RBAC — Role-Based Access Control — permissions attach to a job role, not an individual person. A 'Billing Clerk' role can view invoices but not delete records. This is the most common model in business systems today. Simple way to remember it: DAC, the owner decides. MAC, the system decides. RBAC, the role decides.

DELIVERY CUE

Use three fingers, one per model, as you introduce each — a small physical anchor helps students recall three distinct things instead of blurring them together. Land firmly on the 'owner / system / role' recap line — that's the exam-ready sentence.

SLIDE 7

Authentication & Availability · *suggested time ~1.5 min*

SCRIPT

Two more pieces complete the picture. Authentication verifies a user IS who they claim to be — before access control even applies. That's passwords, biometrics like fingerprint or face recognition, and multi-factor authentication. And availability — access control means nothing if the system is down when someone legitimately needs the data. That's replication and failover systems, regularly tested backups, and load balancing.

DELIVERY CUE

Speak the final tie-together sentence slowly and clearly — it's the single most quotable line on this slide, and a strong candidate for a quiz or exam question.

SLIDE 8

Case Study — Cell C Ransomware Breach (2024) · *suggested time ~2.5 min*

SCRIPT

Let's ground this in a real South African breach. In late 2024, attackers gained unauthorised access to internal Cell C systems and exfiltrated personal, contact, and financial data — ID numbers, banking details, SIM metadata. The breach was later confirmed when roughly 2 terabytes of that data appeared on the dark web, linked to the RansomHouse group, potentially exposing 7.7 million subscribers. So what does this have to do with what we just covered? Investigators pointed to insufficient network segmentation — sensitive customer data wasn't isolated behind stricter, need-to-know access boundaries. That let a single compromised entry point reach a massive dataset. Proper RBAC combined with network segmentation — least privilege applied at the data layer — could have limited how far that one compromised credential was able to travel.

DELIVERY CUE

This is your first case study — deliberately slow down here. Case studies are where attention typically peaks, so don't rush to the next slide. Pause after 'proper RBAC... could have limited how far' to let students connect it back to slide 6 and 7.

13–21 MIN · OUTCOME 2 — PROTECTION OF SENSITIVE DATA

SLIDE 9

Outcome 2 Divider · *suggested time ~20 sec*

SCRIPT

Outcome 2: protection of sensitive data. Not all data is equally dangerous if it's exposed — and, as we'll see, some danger only appears once pieces of data are combined.

What Makes Data 'Sensitive'? · *suggested time ~2 min*

SCRIPT

Sensitivity isn't one category — there are four ways data can become sensitive. Inherently sensitive: always confidential by nature, like ID numbers, banking PINs, passwords. Sensitive by source: drawn from a confidential system, like intelligence reports or patient medical records. Sensitive in context: harmless alone, but revealing when combined — a birth date plus a postal code can identify a specific

person. And temporal sensitivity: harmless right now, but sensitive until a certain date — think unreleased financial results or an unannounced product launch. Only that first category is obviously sensitive. The 'sensitive in context' category is where things get interesting — and that's exactly where the next slide picks up.

DELIVERY CUE

This slide sets up the next one directly — say the last sentence as a genuine transition, not just a summary, so the class expects what's coming.

SLIDE 10

The Inference Problem · *suggested time ~2.5 min*

SCRIPT

Occurs when a user can deduce sensitive information from data that isn't itself classified as sensitive — which undermines security even when direct access is properly blocked. There are five ways this happens. Direct inference — querying and retrieving sensitive data outright because access control was weak. Aggregation — combining several small, non-sensitive facts to reveal a sensitive whole. Negative result — inferring something from the absence of expected data. Existence testing — simply determining whether a record exists at all, like 'is this person in the database?' And probabilistic inference — using statistics to make a highly confident guess at a sensitive value, even without certainty.

SLIDE 11

Defending Against Inference · *suggested time ~2 min*

SCRIPT

So how do we defend against this? Four techniques. Data suppression — remove rare or identifying results entirely; think census data where small household counts are rounded or suppressed so individuals can't be re-identified. Data modification — add statistical noise or swap values to obscure patterns. Query restrictions — limit the number, type, or complexity of queries any one user can run. And view-based access control — expose only aggregated or masked views, never raw tables.

DELIVERY CUE

The trade-off sentence is the exam-relevant takeaway for this whole outcome — say it deliberately, maybe even repeat it: 'perfect privacy and perfect data utility can't coexist.'

SLIDE 14

OUTCOME 3 — DATA MINING & BIG DATA

SLIDE 15

Outcome 3 Divider · *suggested time ~20 sec*

SCRIPT

Outcome three, our last one: data mining and Big Data. This is about turning huge datasets into insight — and, at the same time, into a much bigger attack surface.

DELIVERY CUE

Data Mining: Insight & Risk · *suggested time ~2 min*

SCRIPT

Data mining means analysing large datasets with statistics and machine learning to uncover patterns that aren't immediately obvious. Used well, it's genuinely valuable: fraud detection spots unusual transactions before real damage occurs; intrusion detection uses anomaly recognition to catch cyberattacks early. But the same technique cuts both ways. Privacy breaches happen when anonymised datasets are linked together to re-identify real people. Insider misuse happens when authorised staff mine data beyond what their job actually requires. And function creep happens when data collected for one stated purpose quietly gets reused for another. As the textbook frames it, the real challenge is using data mining for defence without it becoming a tool for exploitation.

SLIDE 12

Big Data: The 5 V's · *suggested time ~2 min*

SCRIPT

Big Data is usually defined by five characteristics — the 5 V's. Volume: the massive quantity of data generated every day. Velocity: the speed at which it's created and processed, often in real time. Variety: everything from structured tables to unstructured text, images, and video. Veracity: the accuracy, quality, and trustworthiness of the data — And Value: the actionable insight and business benefit you can extract from it. Here's the important connection: all five of these simultaneously increase the value of the data — and the difficulty of securing it. More volume, more speed, more variety, more distribution — all of it makes the data more useful and, at the same time, harder to protect. That tension is exactly what the next slide addresses.

SLIDE 13

Securing Big Data Environments · *suggested time ~2 min*

SCRIPT

So how do you actually secure an environment with those five characteristics? Four practical answers. Distributed access control — making sure only authorised users can reach data that's spread across many clustered servers, not just one central database. Encryption in transit and at rest — protecting data from interception whether it's moving over the network or sitting on disk. Privacy-preserving analytics — techniques like differential privacy that let you run useful analysis without ever exposing the raw sensitive data itself. And real-time security monitoring — detecting and responding to threats inside high-speed, high-volume data streams, where traditional periodic auditing is too slow. Notice these four challenges are really Outcomes 1 and 2 — access control and sensitive data protection — scaled up to a distributed, high-volume environment.

DELIVERY CUE

Explicitly call back to Outcome 1 and Outcome 2 here — this is the moment the whole session ties together into one coherent story rather than three separate topics.

29–35 MIN · RECAP, QUIZ & CLOSE

SLIDE 20

Recap — Key Takeaways · *suggested time ~1.5 min*

SCRIPT

Let's bring it all together. Outcome one, access control: DAC, MAC, and RBAC — paired with authentication and auditability — decide and prove who can touch data. Outcome two, sensitive data: classify it properly, watch for inference — aggregation, existence testing, and the rest — and balance security against precision. Outcome three, data mining and Big Data: powerful for fraud and intrusion detection, but volume and distribution multiply the attack surface. Any questions on any of those three before we do a quick quiz?

DELIVERY CUE

Genuinely pause and scan the room for hands here — don't treat this as rhetorical. This is your real checkpoint before the quiz; a hand raised now saves confusion during the quiz.

SLIDE 21

Quiz Time — Quick-Fire Recap Quiz · *suggested time ~4 min*

SCRIPT

Five questions, one per outcome area — let's run through them. You can shout out answers, or if your class has Kahoot set up, I'll switch to that now. [Read each question and its four options aloud, pause for responses, then reveal the answer before moving to the next.] Question one: which access control model assigns permissions based on job function — DAC, MAC, RBAC, or ACL? [Pause. Answer: C, RBAC.] Question two: combining several non-sensitive facts to reveal a sensitive one is called suppression, aggregation, encryption, or replication? [Pause. Answer: B, Aggregation.] Question three: which of the 5 V's refers to data trustworthiness and accuracy — Volume, Velocity, Veracity, or Value? [Pause. Answer: C, Veracity.] Question four: a birth date plus a postal code revealing someone's identity is an example of direct inference, sensitive-in-context data, MAC, or data mining? [Pause. Answer: B, Sensitive-in-context data.] Question five: what does view-based access control protect against — ransomware, inference attacks, power outages, or slow queries? [Pause. Answer: B, Inference attacks.]

DELIVERY CUE

Cover the answer column on your own screen if presenting live from the laptop with notes visible, or advance one build/click at a time so students can't just read ahead. Genuinely wait 5–10 seconds after each question before revealing — resist the urge to fill the silence.

SLIDE 22

References · *suggested time ~30 sec*

SCRIPT

All sources for today's stats and case studies are listed here — the core textbook, Pfleeger and Pfleeger's Security in Computing, plus independent breach reports and news coverage so the case studies are grounded in more than one source each. Feel free to photograph this slide.

DELIVERY CUE

Don't read every citation aloud — name the textbook, gesture at the rest, and move on. This slide is for their notes, not for narration.

SLIDE 23

Thank You · *suggested time ~30 sec*

SCRIPT

That's Study Unit 2 — access control, sensitive data protection, and data mining and Big Data. Happy to take any questions on any of the three.

DELIVERY CUE

Stop talking once you've said this — let the silence invite questions rather than filling it yourself. Stay standing at the front rather than starting to pack up, which visibly signals 'I'm still available.'

Presenting This to a University Class

Practical guidance for delivering this 35-minute, 23-slide session to an ITRI625 class — pacing, engagement, and handling the live quiz.

Before You Start

- **Time yourself once, out loud.** Run through the full script once at a normal speaking pace before class, with a timer running. 35 minutes of slide content usually takes 40+ minutes the first time you say it out loud — trim, don't rush, if you're over.
- **Know your case studies cold.** The Cell C, National Public Data, and Salesforce/SaaS breaches (slides 9, 14, 19) are what will actually stick with students. Be ready to answer a follow-up question on any of them without looking at the slide.
- **Decide on Kahoot before you walk in.** Slide 21 works either live via kahoot.com or as a simple show-of-hands. If you're using Kahoot, load the 5 questions in beforehand and test the join code on the classroom Wi-Fi — don't discover a connectivity issue mid-quiz.
- **Print or screen the notes separately.** Don't present from a laptop screen that the front row can see your speaker notes on — use presenter view, a printed copy of this script, or a second screen.

Pacing the 35 Minutes

- **Protect the case studies, not the checklists.** If you're running behind, trim detail from the checklist-heavy slides (6, 8, 13, 18) rather than cutting a case study short — the real-world examples are what make the theory memorable.
- **Use the roadmap slide as a live checkpoint.** Glance at your watch when you move into each outcome. If you're not starting Outcome 2 by roughly the 13-minute mark, you're behind — pick up the pace on Outcome 1's remaining slides rather than the later ones.
- **Build in two deliberate pauses.** After the Cell C case study (slide 9) and after the inference problem (slide 12) — both are conceptually dense. A 3–5 second pause and a 'does that make sense?' does more for retention than pushing straight through.

Keeping the Room Engaged

- **Ask, don't just tell, on the case studies.** Before revealing the 'access-control lesson' on slide 9, try asking the class: 'given what we just covered on RBAC — what do you think went wrong at Cell C?' A few seconds of genuine guessing beats a straight read-through.
- **Use the numbers as hooks, not wallpaper.** Statistics like R2.2bn or 2.9 billion records land better if you pause after saying them than if you read straight through to the next sentence. Let the number sit for a beat.
- **Physical anchors help with lists of three or five.** Counting on your fingers for the three access-control models (slide 7) or the five V's (slide 17) gives visual learners something to track alongside your voice — small, but it measurably helps recall in a 35-minute lecture-style session.

- **Cold-call sparingly, but do it once.** One well-placed direct question — e.g. 'can anyone give me an example of sensitive-in-context data, other than the one on the slide?' — keeps the room alert without turning the whole session into a Q&A.

Running the Quiz (Slide 21)

- **Genuinely pause after each question.** Wait 5–10 seconds minimum before revealing the answer, even if the room is quiet. Silence feels longer to the presenter than to the audience.
- **Kahoot vs show-of-hands.** Kahoot adds energy and a competitive element but costs 2–3 extra minutes for students to join — budget for that if you use it. Show-of-hands is faster and safer if you're already tight on time.
- **Briefly justify each answer, not just state it.** After revealing 'C — RBAC' for question 1, add one sentence tying it back to the slide it came from ('role decides — that's RBAC') so the quiz reinforces the lecture instead of feeling disconnected from it.

Voice, Body Language & the Room

- **Vary your tone at the three section breaks.** Slides 5, 10, and 15 mark new outcomes — a small deliberate shift in pace or energy at each one helps the class mentally 'file' the session into three parts rather than one long stream.
- **Move, but don't pace nervously.** Step toward the class (away from the podium/screen) during the case-study slides where you're not reading dense text — it signals 'this part is the important part, listen up.'
- **Don't apologise for slide density.** Slides 6, 8, 12, 13 and 18 pack in a lot of content by design. Rather than saying 'sorry, this slide is busy,' group the content into two or three spoken ideas (as scripted above) and move through it with confidence.

Handling Questions

- **Anticipate a POPIA question.** Slide 6 mentions POPIA compliance directly — be ready for a question on how POPIA specifically relates to access control and auditing if your class has covered data protection law elsewhere in the programme.
- **Anticipate a 'why not just encrypt everything' question.** This is a natural question after slide 13's security-vs-precision trade-off. Short answer: encryption protects data at rest/in transit, but doesn't stop authorised users from making valid queries that, combined, still leak information — that's the inference problem, and it's a separate defence layer.
- **If you don't know, say so and move on.** With five independent breach sources cited (slide 22), a student may ask for a detail beyond what's on the slide. It's fine to say 'good question — I'd need to check the source report for that level of detail' rather than guessing.